

Insuline and related drugs

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◉ **Poypeptide Hormones**

- Peptide hormones are composed of amino acids of certain number and sequence, which is determined by genetic information in the DNA

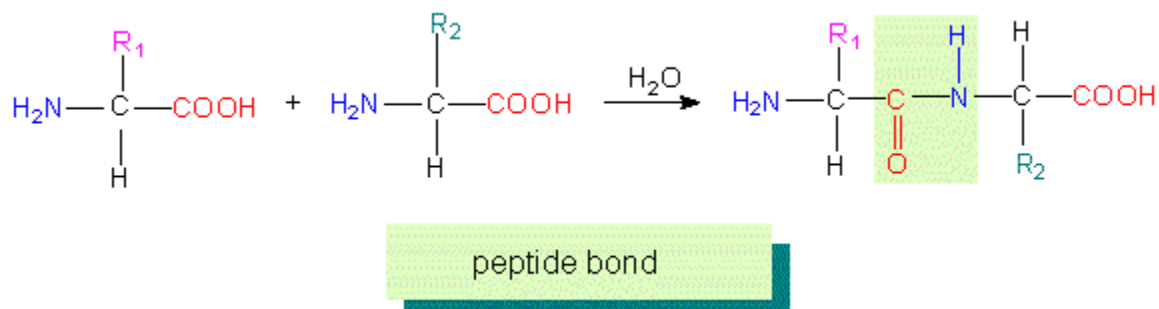
◉ **Peptides include:**

- Dipeptides (2 aminoacide residue)
- Tripeptide (3 aminoacide Residue)
- oligopeptides (10-30 aa residue)
- polypeptides (30-50 aa residue)
- proteine (greater than 50 aminoacides residue)
- Glycoproteine (protein covalently bonded with a carbohydrate).

- ◉ Proteins, growth factors and peptide hormones are synthesized in the endocrine gland as large precursor molecule, **prohormones**.
 - the latter contains a leader sequence of amino acids
 - Enzymatic cleavage of preprohormones yields prohormones from which hormones are liberated.

◉ Chemistry

- Peptide hormones are biosynthesised from L-aminoacides).
- The peptide or amide bond is formed by linking of the a COOH group of one amino acids to the NH₂ group of the other with elimination of H₂O.



Natural aminoacides and, their 3 & 1 letter designation

Amino Acid	Three-Letter Abbreviation	One-Letter Abbreviation
Alanine	Ala	A
Arginine	Arg	R
Asparagine	Asn	N
Aspartic acid	Asp	D
Cysteine	Cys	C
Glutamic acid	Glu	E
Glutamine	Gln	Q
Glycine	Gly	G
Histidine	His	H
Isoleucine	Ile	I
Leucine	Leu	L
Lysine	Lys	K
Methionine	Met	M
Phenylalanine	Phe	F
Proline	Pro	P
Serine	Ser	S
Threonine	Thr	T
Tryptophan	Trp	W
Tyrosine	Tyr	Y
Valine	Val	V

Therapeutic uses of polypeptide hormones:

- ◉ Life time replacement therapy
 - Eg. Insulin
- ◉ Acute or long term stimulation of endocrine axes
 - E.g. Gonadotropines -to stimulate ovarian follicles for ovulations

- ◉ Anticancer uses :
 - Eg.interferones and antibodies
- ◉ Blood volume replacement
 - Albumine and plasma proteins
- ◉ Immunization [Active or Passive]
- ◉ Miscellanouse
 - Trypsine and pepsin

EXAMPLES OF SOME HORMONAL DRUGS

Pancreatic Hormones

⦿ Produces:

- Glucagon from α - cells of Langerhan's $\Rightarrow$ antiinsulin activity
- Insulin from β - cells of Langerhan's $\Rightarrow$ hypoglycemic activity

⦿ Insulin

- Is polypeptide hormone
- Synthesized from pro insulin which is 86 aa precursor
- Pro insulin has negligible hormonal activity

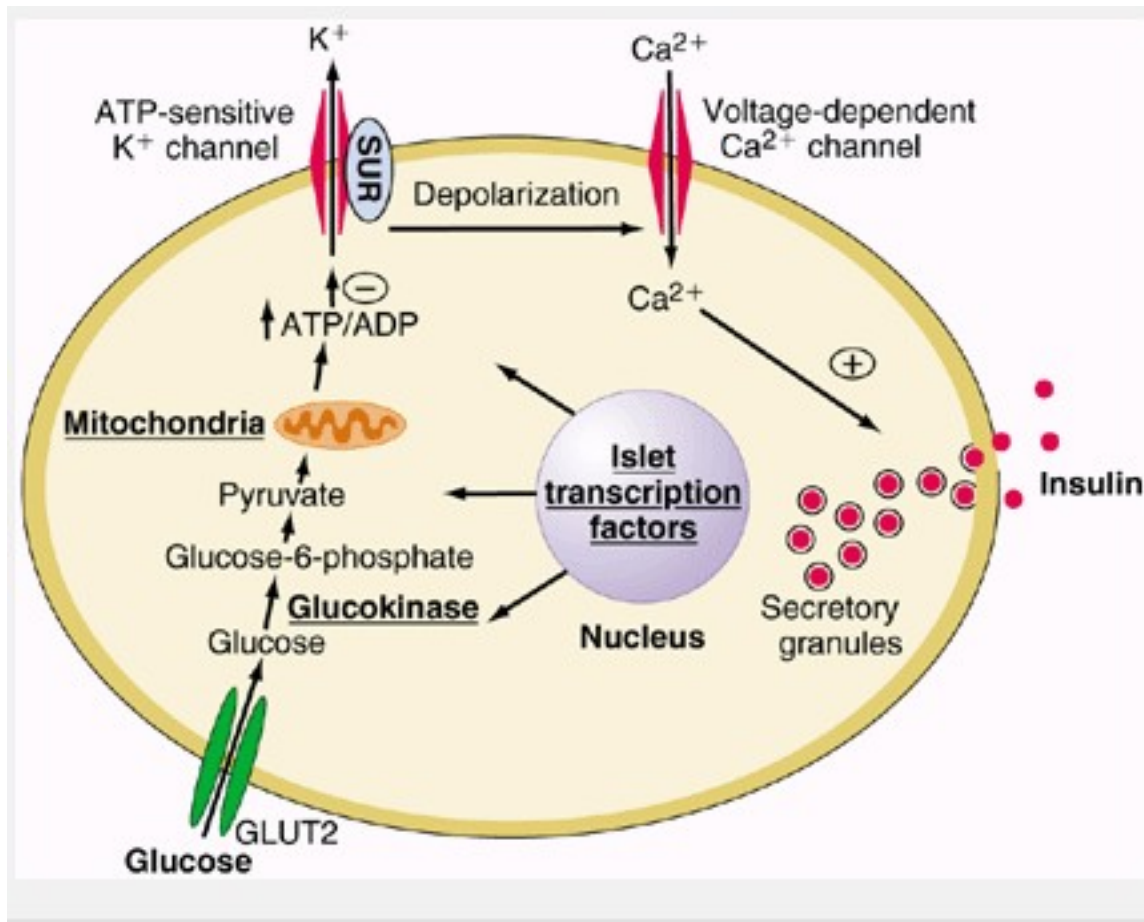


Figure - glucose-stimulated insulin secretion of beta cells of langerhan's

Chemistry

- ◉ Insulin consists of 2 peptide chains:
 - Acidic A chain of 21 aa residues
 - Basic B chain of 30 aa residues
- ◉ It has 3 disulphide bridges b/n aa residues
 - A7 - B7
 - A20 - B19
 - A6 - A11
- ◉ In addition to the disulphide linkage the two chains are held together by
 - Hydrophobic bonds
 - Hydrogen bonds
 - Salt linkage

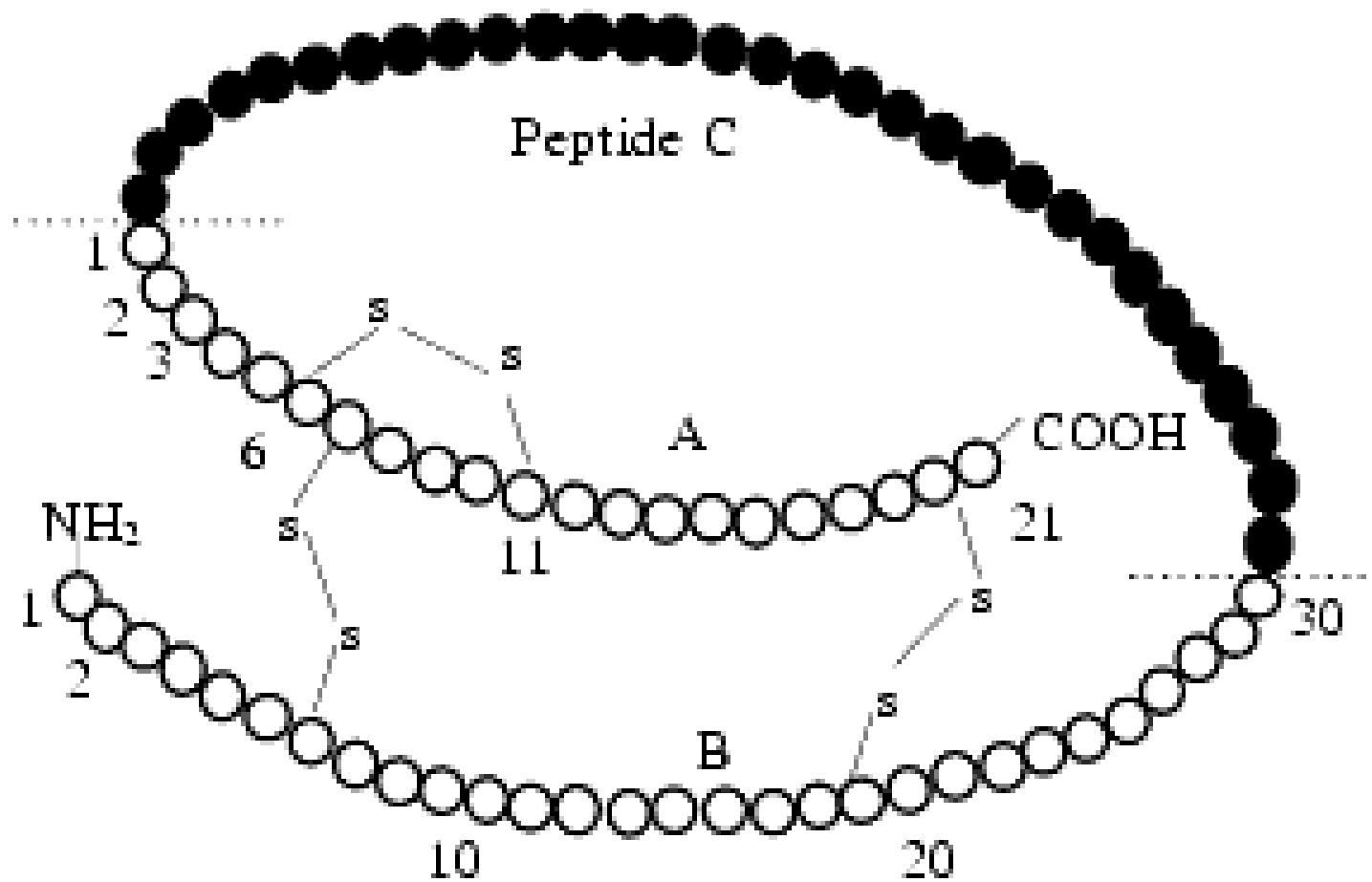
Species variation

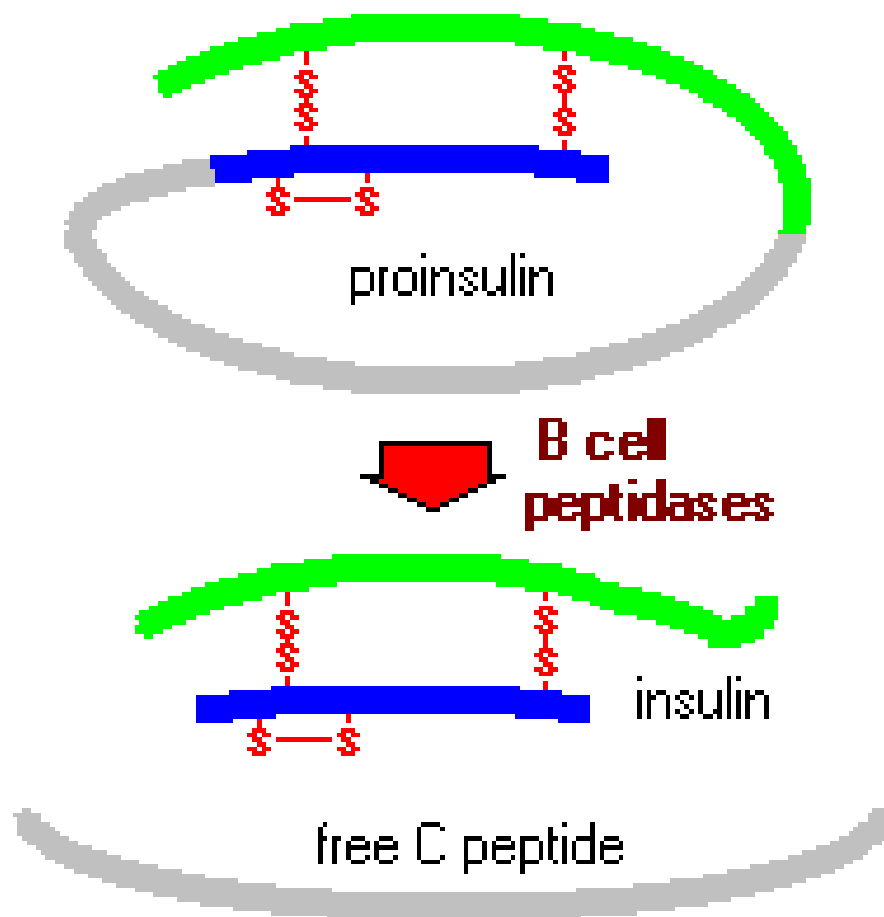
- ◉ Same structural framework but variable aa substitution
- ◉ Porcine insulin is the closest to human insulin
- ◉ Differences of insulin from different spps

<u>Species</u>	<u>A8</u>	<u>A10</u>	<u>B28</u>	<u>B29</u>	<u>B30</u>
Human	Thr	Ile	Pro	Lys	Thr
Lispro	Thr	Ile	Lys	Pro	Thr
Porcine	Thr	Ile	Pro	Lys	Ala
Bovine	Ala	Val	Pro	Lys	Ala

SAR:

1. significant loss of hormonal activity is observe up on removal of aa units from chain A
 2. Activity is reduced to 10 - 20% by replacement of A1 glycine with L-Ala but fully active up on replacing by D-Ala.
 3. Several aa residues of B are not essential. Up to the first 6 and the last 3 can be removed.
 4. Breakage of disulphide linkage results in loss of activity.
- **** Being protein insulin cannot be given orally as it is digested by proteases and other enzymes.





CHEMICAL MODIFICATION OF INSULIN

- ◉ Synthesis
- ◉ Semisynthesis
- ◉ Recombinant DNA

■ Synthesis

- Total synthesis of human insulin was laborious and totally difficult
- Require more than 100 steps
- Synthetic insulin obtained in three steps
 - Synthesis of chain A
 - Synthesis of chain B
 - Linking of the two by disulphide linkage

◉ Semi-synthesis

By coupling parts of native hormone with synthetic peptide.

E.g. conversion of porcine insulin (B30=Ala) to human insulin (B30=thr) by enzymatic removal of sequence B23-30 and coupling with octapeptide of human insulin

◉ Recombinant DNA

1st method- insertion of genes for production of either chain A or chain B into a special strain of E.coli & then combining the 2 chains chemically to produce insulin w/h is structurally and chemically similar to human insulin

- ⊙ **2nd method-** insertion of gene for the entire proinsulin molecule into special *E. coli* cells that are then grown by fermentation process
 - The connecting C- peptide is then enzymatically cleaved from proinsulin
- ⊙ Human insulin produced by rDNA technology is less antigenic than that produced from animal sources
- Lispro - is insulin obtained through r-DNA

⦿ Insulin Preparations

1. Rapid acting
2. Intermediate acting
3. Long acting

⦿ * Dimerization of insulin occurs by complexing with Zn ion

- ◉ DM- metabolic disorder characterized by an increase in blood glucose
- ◉ Results from inadequate insulin secretion, insulin resistance
- ◉ Types of DM
 - Type I (IDDM)
 - Type II (NIDDM)

⦿ Insulin preparations

- Rapid acting
- Intermediate acting
- Long acting

- Zn- usually found in insulin pharmaceutical formulations to promote stability
- Insulin replacement therapy (IRT) is the only efficacious treatment for type I DM
- Side effect- hypoglycemia

ORAL HYPOGLYCEMIC AGENTS

- Some milder forms of diabetes mellitus that do not respond to diet management or weight loss and exercise can be treated with **oral hypoglycemic** agents.
- The success of oral hypoglycemic drug therapy is usually based on a **restoration of normal blood glucose levels** and the absence of **glycosuria**.

DRUG CLASSES

- ⦿ 1. Sulfonylureas
- ⦿ 2. Biguanides
- ⦿ 3. Alpha glucosidase inhibitors
- ⦿ 4. Thiazolidinediones

Sulfonylureas

- ◉ Sulfonylureas are **the most widely prescribed** drugs in the treatment of **type II** diabetes mellitus.
- ◉ The initial sulfonylureas were introduced nearly 50 years ago and were derivatives of the antibacterial sulfonamides.
- ◉ Although their structural similarities to the sulfonamide antibacterial agents are readily apparent, the sulfonylureas possess no antibacterial activity.

Mechanism of Action

- ⦿ The primary mechanism of action of the sulfonylureas is *direct stimulation of insulin release from the pancreatic β -cells*.
- ⦿ At higher doses, these drugs **also decrease hepatic glucose production**
- ⦿ The second-generation sulfonylureas may possess additional extra-pancreatic effects that **increase insulin sensitivity**

- ◉ The sulfonylureas are **ineffective** for the management of type I and severe type II diabetes mellitus,
 - since the number of viable β -cells in these forms of diabetes is small.
- ◉ Severely obese diabetics often respond poorly to the sulfonylureas, possibly **because of the insulin resistance that often accompanies obesity.**

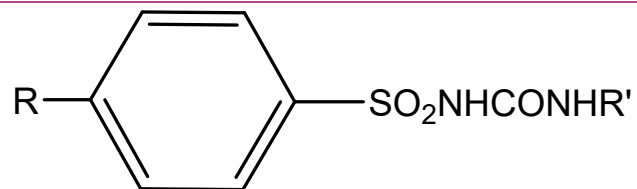
◉ Sub classified as

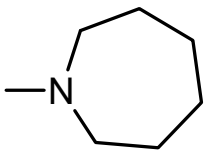
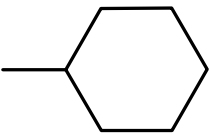
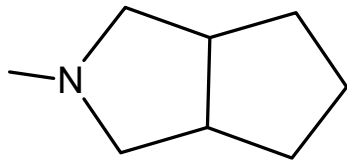
- 1st generation

- 2nd generation

A. 1ST GENERATION

- ◉ The first-generation sulfonylureas are **not frequently used in the modern management of diabetes mellitus** because of their
 - relatively low specificity of action
 - delay in time of onset
 - occasional long duration of action, and
 - a variety of side effects
- ◉ They also tend to have more adverse drug interactions than the second-generation sulfonylureas



	R	R'
Tolbutamide	CH ₃	n-Butyl
Chlorpropamide	Cl	n- Propyl
Tolazamide	CH ₃	
Acetohexamide	CH ₃ CO	
Gliclazide	CH ₃	

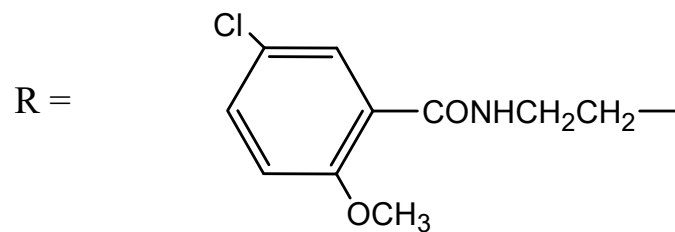
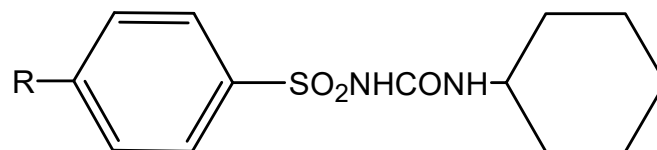
SAR

- These are **urea** derivatives with an aryl sulfonyl group in the 1 position and an aliphatic group in the 3 position
- Small alkyl substituents such as methyl or ethyl are not active
- In 1st generation, the aromatic substituent is relatively simple (e.g. methyl, amino, acetyl, chloro, bromo, methylthio, CF₃) however, 2nd generation analogues have a larger p-(β -arylcarboxyamidoethyl) group that leads to higher potency.

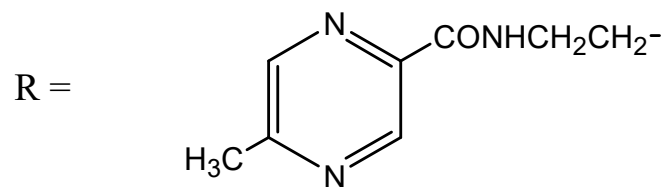
- The R' confers lipophilic property to the molecule
- Aryl substituents at this position gives toxic compounds

B. 2ND GENERATION

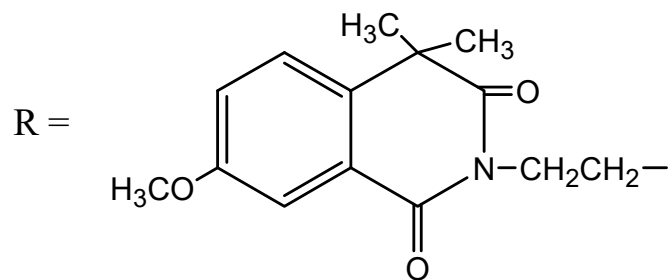
- The second-generation sulfonylureas display
 - A higher specificity and affinity for the sulfonylurea receptor
 - More predictable pharmacokinetics in terms of time of onset and duration of action
 - Fewer side effects.



Glibenclamide

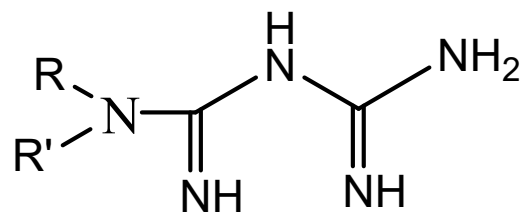


Glipizide



Glipizide

2) BIGAUNIDES



	R	R'
Phenformin	H	PhCH ₂ CH ₂
Metformin	CH ₃	CH ₃
Buformin	H	n-Butyl

◉ MOA

- reducing sugar absorption from the gastrointestinal tract.
 - decrease gluconeogenesis while increasing glucose uptake by muscle and fat cells
- ◉ Unlike the sulfonylureas, these are not hypoglycemic agents but rather act as **antihyperglycemics**
- ◉ **Metformin** is approved for the treatment of patients with NIDDM that are refractory to dietary management alone

3) *ALPHA-GLUCOSIDASE INHIBITORS*

- ◉ The α -glucosidase inhibitors **primarily act to decrease postprandial hyperglycemia** by *slowing the rate at which* carbohydrates are absorbed from the gastrointestinal tract.
- ◉ Work by preventing digestion of carbohydrates
- ◉ Alpha glucosidase inhibitors are saccharides w/h act as competitive inhibitors of enzymes needed to digest carbohydrates
- ◉ Oligosaccharides $\rightarrow$ glucose by alpha glucosidase
- ◉ Inhibition of this enzyme reduces the rate of digestion of carbohydrates

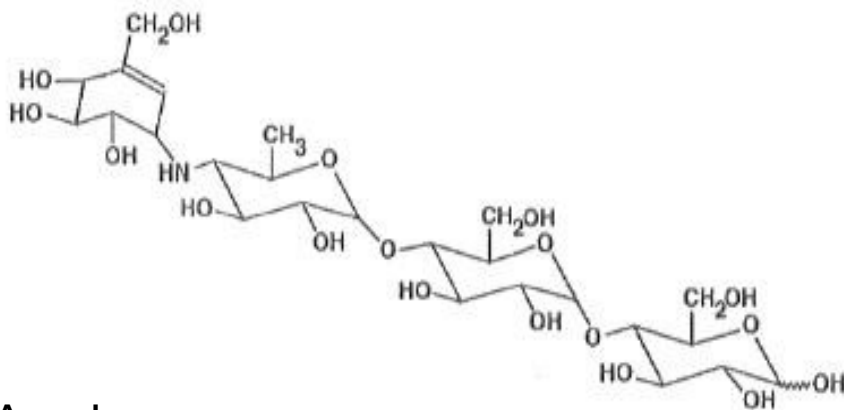
- ⦿ The prolongation of the intestinal absorption of carbohydrates results keeping postprandial hyperglycemia under control

i. Acarbose

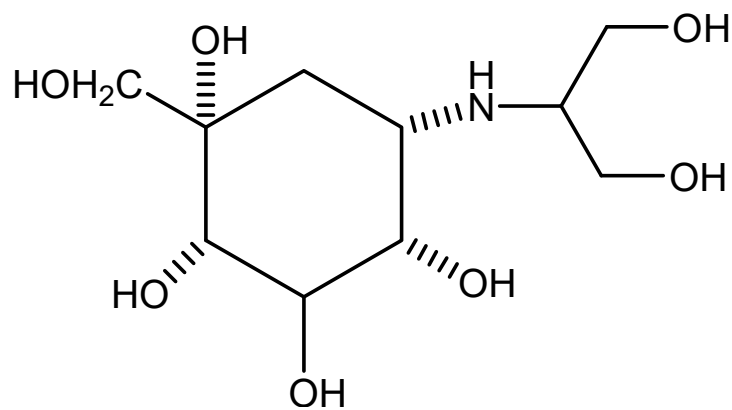
- is a complex oligosaccharide
- it delays CHO metabolism by inhibiting α -D-Glucosidase

ii. voglibose

- ⦿ is orally active inhibitor of α -D-Glucosidase
- ⦿ also used for other CHO dependant disorders
- ⦿ much more potent and with fewer side effects than Acarbose



Acarbose

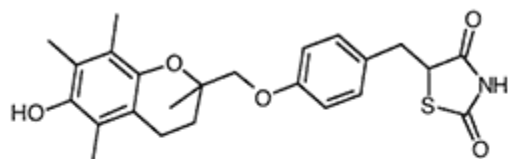


Voglibose

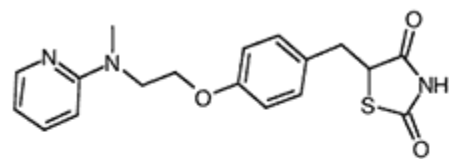
4) THIAZOLIDINEDIONES

- ⦿ Thiazolidinediones (sometimes termed **glitazones**) are a novel class of drugs that were initially identified **for their insulin-sensitizing properties**.
- ⦿ They all act to decrease insulin resistance and enhance insulin action in target tissues.
- ⦿ Thiazolidinediones activate the nuclear peroxisome proliferator-activated receptor (PPAR) γ and modulate the expression of insulin-sensitive genes

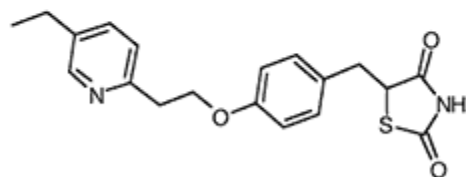
- ⦿ The patient who would benefit the most from a thiazolidinedione is
 - a type II diabetic with a substantial amount of insulin resistance



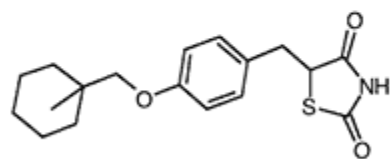
Troglitazone



Rosiglitazone



Pioglitazone



Ciglitazone

WAITING FOR EXAM?

